

Joint Control Barrier Function for Target Visibility and Thrust Deliverability

A camera-plane QP over the commanded inertial acceleration

Soft–Precise–Landing

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Scope and notation. This note follows the notation of the manuscript’s Visibility CBF subsection and Implementation Appendix verbatim (frames $\mathcal{I}/\mathcal{C}$, feature points $\tilde{\mathbf{r}}_i$, coupling $\tilde{L}_{\omega,i} = L_{\omega}M$, lean $\boldsymbol{\theta}$, cap θ_{cap} , drift $\mathbf{d}_i$, look-ahead τ) and extends it with one additional symbol, A_{cap} , the vehicle’s measured maximum thrust-to-mass ratio. Visibility is fundamentally an *attitude* constraint (the body tilt shifts the FoV), but the attitude actually commanded is a byproduct of the desired inertial acceleration ${}^{\mathcal{I}}\mathbf{a}_d$ the outer loop produces. Rather than solving the tilt-only projection and separately deciding how much vertical thrust to spend, this implementation solves one projection directly over ${}^{\mathcal{I}}\mathbf{a}_d$ that jointly enforces (i) the barrier condition of the manuscript’s Visibility Forward Invariance theorem and (ii) the deliverable-lateral-acceleration bound of the manuscript’s Property 1, in a single iteration rather than a tilt projection followed by a separate deliverability cap. The coupling itself is unchanged: the **rotational part of the IBVS interaction matrix**, L_{ω} , evaluated at the measured image point — exact, *depth-independent*, no de-rotation or virtual frame required.

Notation

Symbol	Meaning
$\tilde{\mathbf{r}}_i, \tilde{r}_{i,k}$	measured camera-frame feature point i , tangent units, per axis k
$L_\omega(\tilde{\mathbf{r}}_i)$	rotational interaction matrix (roll/pitch columns), $\partial \dot{\tilde{\mathbf{r}}}_i / \partial \boldsymbol{\omega}_{\text{rp}}$
$\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the <i>rotation-axis</i> vector)
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	lean→rotation map: $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (lean $+x$ is a rotation about $+y$)
$\tilde{L}_{\omega,i} = L_\omega M$	effective tilt coupling (orthogonal M , so row norms are preserved)
$\mathcal{I} \mathbf{a}_{\text{d}}$	desired inertial acceleration; this QP’s decision variable
$\mathcal{I} a_{\text{d},z}$	desired vertical acceleration component ($a_z \equiv \mathcal{I} a_{\text{d},z} $)
$\boldsymbol{\theta} = R_z(-\psi_{\text{b}})(\mathcal{I} \mathbf{a}_{\text{d},xy} / \mathcal{I} a_{\text{d},z})$	<i>derived</i> lean-direction vector, used to build the commanded attitude
$\boldsymbol{\theta}_{\text{curr}}$	current tilt, from the realized attitude
A_{cap}	vehicle’s maximum achievable thrust-to-mass ratio (measured)
θ_{cap}	deliverable-tilt cap (60° nominal; a 2×-hover-thrust margin)
$\mathbf{d}_i$	exogenous (translation) feature drift, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\varphi_{\text{max}} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
$\delta, \dot{\delta}$	primary-marker half-spread (loom rate)
τ, g	drift look-ahead; gravitational acceleration

1 Barrier on the real camera plane

Per image axis k , for feature point i ,

$$\boxed{h_{i,k} = \varphi_{\text{max},k} - |\tilde{r}_{i,k}|}. \quad (1)$$

$h_{i,k} \geq 0 \iff$ feature point i is on the sensor on axis k . $\tilde{\mathbf{r}}_i$ is the *measured* camera-frame feature point — nothing is de-rotated. This implementation tracks a single feature point (the target’s decoded centroid), i.e. $N = 1$ in the manuscript’s general per-feature barrier.

2 Tilt→feature coupling, expressed in acceleration units

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point:

$$\dot{\tilde{\mathbf{r}}}_i = L_\omega(\tilde{\mathbf{r}}_i) \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}_i, \quad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}_{(x,y)=\tilde{\mathbf{r}}_i}. \quad (2)$$

L_ω contains **no depth**. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$ (using the true gyro rate), so it carries only genuine target motion, not our own tilting.

A commanded lean produces a rotation about the perpendicular axis, $\boldsymbol{\omega} = M\boldsymbol{\theta}$, and the lean is itself the direction of the commanded lateral acceleration relative to the commanded vertical acceleration, $\boldsymbol{\theta} = R_z(-\psi_b)(\mathcal{I}\mathbf{a}_{d,xy}/\mathcal{I}a_{d,z})$ (the manuscript’s own definition, with ψ_b composed here with the fixed camera-mount alignment rotation this platform’s downward-mounted camera requires; the composition reduces to $R_z(-\psi_b)$ alone for an ideally body-aligned camera). Writing P for this composed rotation and substituting both relations into (2) gives the feature velocity directly as a function of the acceleration command:

$$\boldsymbol{\omega} = M\boldsymbol{\theta}, \quad \tilde{L}_{\omega,i}\boldsymbol{\theta} = \underbrace{\left(\frac{\tilde{L}_{\omega,i}P}{a_z}\right)}_{N_i(a_z)} \mathcal{I}\mathbf{a}_{d,xy}. \quad (3)$$

$N_i(a_z) = \tilde{L}_{\omega,i}P/a_z$ is the barrier’s Jacobian with respect to the lateral acceleration directly. It depends on a_z — the same quantity the joint projection below also solves for — which is exactly the coupling that projection is built to resolve, rather than treating a_z as an external given. For a finite acceleration change the exact map is the rotation homography of the measured point; the first-order form

$$\tilde{\mathbf{r}}_i(\mathcal{I}\mathbf{a}_{d,xy}) \approx \tilde{\mathbf{r}}_i + N_i(a_z)\mathcal{I}\mathbf{a}_{d,xy} - \tilde{L}_{\omega,i}\boldsymbol{\theta}_{\text{curr}} \quad (4)$$

is the local linearization, accurate over one control step. $\boldsymbol{\theta}_{\text{curr}}$ is read directly from the realized attitude (not from $\mathcal{I}\mathbf{a}_d$), so it needs no rotation from acceleration units.

3 The visibility–deliverability projection

Two physical constraints bind the same variable $\mathcal{I}\mathbf{a}_d$: the barrier condition of (1) along the feature response (2)–(3), and the deliverable-lateral-acceleration bound of the manuscript’s Property 1 (inner-loop deliverable-tilt saturation bounds $\|\mathcal{I}\mathbf{a}_{d,xy}\|$ by a state-dependent fraction of $|\mathcal{I}a_{d,z}|$). Writing $\mathcal{I}\mathbf{a}_d^*$ for the conditioned output (paralleling the manuscript’s $\boldsymbol{\theta}^*$ notation for its own tilt-only projection),

$$\boxed{\begin{aligned} \mathcal{I}\mathbf{a}_d^* &= \arg \min_{\mathcal{I}\mathbf{a}_d} \|\mathcal{I}\mathbf{a}_d - \mathcal{I}\tilde{\mathbf{a}}_d\|^2 \\ \text{s.t. } &|\tilde{\mathbf{r}}_i + N_i(a_z)\mathcal{I}\mathbf{a}_{d,xy} - \tilde{L}_{\omega,i}\boldsymbol{\theta}_{\text{curr}} + \tau\mathbf{d}_i| \preceq \boldsymbol{\varphi}_{\text{max}}, \quad \|\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}} \end{aligned}} \quad (5)$$

where $\mathcal{I}\tilde{\mathbf{a}}_d$ is the filtered command of the manuscript’s low-pass filter (Implementation Appendix), the unconstrained target this projection stays closest to. The first constraint is the manuscript’s per-feature FoV box (its Eq. 13, Implementation Appendix), now written directly on $\mathcal{I}\mathbf{a}_{d,xy}$; the second is the true thrust-deliverability sphere in gravity-shifted acceleration space (the vehicle’s total thrust vector is $m(\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3)$, and $\|\mathcal{I}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}}$ is exactly the statement that this stays within the measured maximum). Both are convex for a fixed a_z ; since $N_i(a_z)$ itself depends on a_z , the two projections are iterated together (alternating projection onto the box at the current a_z estimate, then onto the sphere, updating a_z for the next pass) until convergence. This is a genuine joint solve for the lateral and vertical acceleration together, not a lateral solve followed by a separate vertical adjustment.

- **Directional** — only the outward-binding box row is active; inward (recovery) and tangential accelerations are free. No strangling, matching the manuscript’s own $\dot{h}_{i,k}$ argument.

- **Envelope-aware** — the deliverability side of (5) is the vehicle’s actual measured thrust ceiling A_{cap} , not θ_{cap} applied *as if* a_z were fixed at hover; it tightens or loosens correctly as a_z moves away from hover.
- **Forward-invariant** — the look-ahead shifts the box anchor by $\tau \mathbf{d}_i$, so the projection accounts for centroid drift over the next τ s in both phases (Sect. 4), exactly as in the manuscript’s realization.
- **Bounded by construction** — because ${}^{\mathcal{I}}\mathbf{a}_d^*$ always satisfies the sphere constraint, the derived lean $\boldsymbol{\theta}^* = P {}^{\mathcal{I}}\mathbf{a}_{d,xy}^*/a_z^*$ used to build attitude (Sect. 8) satisfies $\|\boldsymbol{\theta}^*\| \leq \tan \theta_{\text{cap}}$ at the solved a_z^* automatically; no separate, hover-assumed cap application is needed.

4 Two-phase δ (central-marker convention)

The designated *central* marker is the target. φ_{max} in (5) is phase-dependent:

Phase 1 — central marker decodes. $\delta_k = 0$, so the box uses the full half-FoV $\varphi_{\text{max}} = \mathcal{R}/(2f)$ (this implementation does not apply the manuscript’s inset margin $\delta_m = 15 \text{ px}/f$; see the note at the end of this section). The barrier is centroid-only: the box spans the entire image plane, and only fires if the *centroid* drifts toward the edge. The central marker is deliberately allowed to grow and overflow as the UAV closes in — its visibility *is* the guarantee; using its large half-extent would fire the barrier early. $\dot{\delta}$ is tracked but excluded from φ_{max} in Phase 1.

Phase 2 — central marker overflows (decode-fail, hysteresis-gated against flicker $\Rightarrow$ touch-down range), matching the manuscript’s “on a marker decode failure the projection falls back to a magnitude clamp $\|{}^{\mathcal{I}}\tilde{\mathbf{a}}_{d,xy}\| \leq |{}^{\mathcal{I}}\tilde{\mathbf{a}}_{d,z}| \tan \theta_{\text{cap}}$.” δ_{eff} **ramps in** over a few frames from the visible-*set* half-extent ($\frac{1}{2}\text{ptp}_k$ over ALL visible markers’ corners, so $|\tilde{\mathbf{r}}_i| + \delta_{\text{eff}}$ is the true outermost feature — not one marker):

$$\varphi_{\text{max},k} \rightarrow \varphi_{\text{max},k} - \delta_{\text{eff},k} - \tau \dot{\delta}_{\text{eff},k}.$$

The ramp (not a step) keeps $h_{i,k}$ from dropping discontinuously at engagement, but the ramp gain is clamped so δ_{eff} never *lags* the measured $\frac{1}{2}\text{ptp}$ — a ramp slower than the fill under-protects. Within Phase 2 the visible markers themselves leave (a mild inner sawtooth); $\dot{\delta}$ is reset across those switches. The sphere constraint of (5) applies unchanged in both phases; the deliverability side of the joint solve does not depend on which phase the visibility side is in.

Implementation: hysteresis threshold = 3 consecutive decode-fails before Phase 2 activates; ramp completes over 5 control cycles ($\approx 120 \text{ ms}$ at 42 Hz). Both $\delta_{\text{eff}} = \alpha \delta_{\text{ref}}$ and $\dot{\delta}_{\text{eff}} = \alpha \dot{\delta}_{\text{ref}}$ are scaled by the same ramp factor $\alpha \in [0, 1]$, where δ_{ref} and $\dot{\delta}_{\text{ref}}$ are the half-extent and loom rate measured at the last successful decode.

Note on δ_m : the manuscript’s inset half-extent $\varphi_{\text{max}} = \mathcal{R}/(2f) - \delta_m$ absorbs the one-step linearization residual and the attitude-tracking residual (the robustness margin $\delta_m \geq \bar{d}$ of the Visibility Forward Invariance theorem). This deployed implementation currently uses the raw $\mathcal{R}/(2f)$ without subtracting δ_m — a known gap against the manuscript’s stated margin, not yet closed in code.

5 Feasibility floor and hand-off

The box constraint alone is **infeasible** iff $\varphi_{\max,k} < 0$ for some k under Phase 2 margins (the marker fills the frame within τ). In Phase 1, $\varphi_{\max} = \mathcal{R}/(2f) > 0$ always, so the box is **always feasible** on its own while the central marker decodes. The sphere constraint is always feasible by construction (it is satisfied by $\mathcal{I}\mathbf{a}_d = \mathbf{0}$). Genuine joint infeasibility — the box demanding more lateral acceleration than A_{cap} can deliver at any achievable a_z — is the acceleration-space counterpart of the manuscript’s feasibility condition

$$\sigma_{\min}(\tilde{L}_{\omega,i}) (\tan \theta_{\text{cap}} - \|\boldsymbol{\theta}_{\text{curr}}\|) \geq \max_{i,k} (|\tilde{r}_{i,k}| - \varphi_{\max,k})_+ + \bar{d} \quad \forall i, \quad (6)$$

evaluated here at the solved a_z^* rather than at a fixed hover-assumed θ_{cap} ; when it fails, the iteration converges to the point on the sphere closest (in the box-projection sense) to feasibility, the same graceful degradation the manuscript’s alternating projection already provides for the box alone.

The perception-floor hand-off is triggered by *central-marker overflow* (Phase 2 activation, hysteresis-gated), not by box infeasibility. On Phase 2 entry, or if $\varphi_{\max,k} < 0$ under Phase 2 margins, raise the perception-floor flag and hand off to the texture-free ring flow.

6 Assumptions (and the practical challenge each carries)

1. **The coupling $\tilde{L}_{\omega,i} = L_{\omega}M$ and its sign convention are calibrated.** The fixed 90° map $\boldsymbol{\omega} = M\boldsymbol{\theta}$ bridges the lean direction and the rotation-axis rate that L_{ω} couples to. Any residual mount-dependent sign of L_{ω} is a one-time check: command a small acceleration and confirm the resulting $\Delta\tilde{\mathbf{r}}_i$ matches $\tilde{L}_{\omega,i}$ in sign before trusting the barrier.
2. **The conditioned acceleration is realized** (the $\text{SO}(3)$ loop tracks the attitude built from $\mathcal{I}\mathbf{a}_d^*$), per Assumption on the timescale separation between the outer and inner loops. *Challenge:* attitude bandwidth $\Rightarrow$ lag; the sphere constraint bounds magnitude, not delay.
3. **The drift $\mathbf{d}_i$ is the translation part** ($\dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_{\omega}\boldsymbol{\omega}^{\text{curr}}$) and $\dot{\delta}$ the loom. *Challenge:* finite differences $\Rightarrow$ filter + binding-axis hysteresis; the disturbance enters as *measured* motion, matching the manuscript’s own treatment.
4. **First-order L_{ω} over the cycle.** Exact is the rotation homography of $\tilde{\mathbf{r}}_i$; the linearization (4) is accurate over one control step.
5. **A_{cap} is the vehicle’s true achievable thrust-to-mass ratio**, measured rather than assumed — this instantiates the manuscript’s Property 1 exactly, using a direct, always-current physical bound in place of the fixed $\theta_{\text{cap}} = 60^\circ$ nominal margin.

7 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the projection (5) contain only the measured image point $\tilde{\mathbf{r}}_i$, δ , $\varphi_{\max} = \mathcal{R}/(2f)$, the body rate, the commanded acceleration, and the vehicle’s own A_{cap} and g . L_{ω} is **depth-independent by construction**, and A_{cap} , g are fixed vehicle/physical constants, not measurements of target depth or altitude — the whole formulation remains manifestly scale- and depth-free, matching the manuscript’s own scale-free design throughout.

8 Implementation (per control cycle)

1. (*once*) Calibrate L_ω 's sign / axis convention (Assumption 1); measure A_{cap} .
2. Measure $\tilde{\mathbf{r}}_i$ (board centroid, homography). Advance the two-phase δ state: if the central marker decodes, reset Phase 2 counters and set $\delta_{\text{eff}} = 0$; otherwise increment the decode-fail counter, and once the hysteresis threshold is reached, ramp δ_{eff} toward the visible markers' $\frac{1}{2}\text{ptp}$.
3. Evaluate $L_\omega(\tilde{\mathbf{r}}_i)$; form the drift $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$ and $\dot{\delta}_{\text{eff}}$ (filtered; zero in Phase 1). Read $\boldsymbol{\theta}_{\text{curr}}$ from the current realized attitude.
4. Two-phase margin per Sect. 4. If $\varphi_{\text{max},k} < 0$ for any k , raise the perception-floor flag and hand off to the ring estimator. Otherwise build the joint projection (5).
5. Starting from ${}^{\mathcal{I}}\tilde{\mathbf{a}}_d$ (the filtered desired acceleration, per the manuscript's low-pass filter), jointly project onto the FoV box (using $N_i(a_z)$ at the current a_z estimate) and the deliverability sphere, iterating until convergence, to obtain ${}^{\mathcal{I}}\mathbf{a}_d^*$.
6. **Build the desired attitude directly from ${}^{\mathcal{I}}\mathbf{a}_d^*$** , mirroring the manuscript's own direct-lean construction. Recover $\boldsymbol{\theta}^* = P {}^{\mathcal{I}}\mathbf{a}_{d,xy}^* / a_z^*$ and set the desired body- z axis $\mathbf{r}_{d,3} = [-R_z(\psi_b)\boldsymbol{\theta}^*, 1]^\top / \|\cdot, 1\|$, passing R_d to the SO(3) inner loop and taking the thrust magnitude ${}^{\mathcal{B}}T_u = m |{}^{\mathcal{I}}\mathbf{a}_{d,z}^*| / R_{33}$ directly from ${}^{\mathcal{I}}\mathbf{a}_d^*$, exactly as the manuscript's own R_d /thrust-magnitude construction. Building attitude from the un-filtered ${}^{\mathcal{I}}\mathbf{a}_d^*$ (rather than from a separately low-pass-filtered command) keeps the barrier's guarantee landing exactly on the commanded attitude, with no smearing from a downstream filter time constant.